

# SAT Math

## Ratios, Rates, Proportions, and Units 1

| Question # | ID       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Answer |
|------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.1        | 3f5398a6 | <p>For a person <math>m</math> miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is <math>k</math> seconds. The ratio of <math>m</math> to <math>k</math> can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?</p> <p>A. 10</p> <p>B. 9</p> <p>C. 6</p> <p>D. 5</p> | D      |
| 1.2        | 000259aa | <hr/> <p>A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?</p> <p>A. 0.057</p> <p>B. 0.729</p> <p>C. 17.5</p> <p>D. 24</p>                                                                                                                                    | C      |
| 1.3        | 312ba47c | <p>In a box of pens, the ratio of black pens to red pens is 8 to 1. There are 40 black pens in the box. How many red pens are in the box?</p> <p>A. 5</p> <p>B. 8</p> <p>C. 40</p> <p>D. 320</p>                                                                                                                                                                                                                                                                             | A      |
| 1.4        | 15617f62 | <p>The population density of Worthington is 290 people per square mile. Worthington has a population of 92,800 people. What is the area, in square miles, of Worthington?</p> <p>A. 102,400</p> <p>B. 93,090</p> <p>C. 320</p> <p>D. 32</p>                                                                                                                                                                                                                                  | C      |

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Question # ID

1.5 be35c117

Answer

A

A wind turbine completes 900 revolutions in 50 minutes. At this rate, how many revolutions per minute does this turbine complete?

- A. 18
- B. 850
- C. 950
- D. 1,400

1.6 3f236a64

D

| $x$ | $y$ |
|-----|-----|
| 1   | 4   |
| 3   | 12  |
| 5   | 20  |
| 40  | $k$ |

In the table above, the ratio of  $y$  to  $x$  for each ordered pair is constant. What is the value of  $k$ ?

- A. 28
- B. 36
- C. 80
- D. 160

1.7 6310adbc

D

The ratio of  $t$  to  $u$  is 1 to 2, and  $t = 10$ .

What is the value of  $u$ ?

- A. 2
- B. 5
- C. 10
- D. 20

1.8 aeeaec96

B

How many yards are equivalent to 612 inches? (1 yard = 36 inches)

- A. 0.059
- B. 17
- C. 576
- D. 22,032

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| Question # | ID       |                                                                                                                                                                                                                                                                                                                                                                                                        | Answer |
|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.9        | e9841407 | <p>Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?</p> <p>A. 1 to 4</p> <p>B. 4 to 1</p> <p>C. 1 to 7</p> <p>D. 7 to 1</p>                                                                                                                                                                                                                   | A      |
| 1.10       | ba62b0b0 | <p>A kangaroo has a mass of 28 kilograms. What is the kangaroo's mass, in <u>grams</u>? (1 kilogram = 1,000 grams)</p> <p>A. 28,000</p> <p>B. 1,028</p> <p>C. 972</p> <p>D. 784</p>                                                                                                                                                                                                                    | A      |
| 1.11       | 24ad9dcb | <p>The weight of an object on Venus is approximately <math>\frac{9}{10}</math> of its weight on Earth. The weight of an object on Jupiter is approximately <math>\frac{23}{10}</math> of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?</p> <p>A. 90</p> <p>B. 111</p> <p>C. 140</p> <p>D. 230</p> | C      |
| 1.12       | d0d9ede4 | <p>How many feet are equivalent to 34 yards? (1 yard = 3 feet)</p>                                                                                                                                                                                                                                                                                                                                     | 102    |

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Question # ID

1.13 06a152cd

Answer

A

To make a bakery's signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

- A. 7.5
- B. 10
- C. 50.5
- D. 120

1.14 3ac09984

A

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00

1.15 99550621

B

Makayla is planning an event in a 5,400-square-foot room. If there should be at least 8 square feet per person, what is the maximum number of people that could attend this event?

- A. 588
- B. 675
- C. 15,274
- D. 43,200

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| Question # | ID       |                                                                                                                                                                                                                | Answer |
|------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.16       | 808f7d6c | <hr/> <p>If <math>t = 4u</math>, which of the following is equivalent to <math>2t</math>?</p> <p>A. <math>8u</math></p> <p>B. <math>2u</math></p> <p>C. <math>u</math></p> <p>D. <math>\frac{1}{2}u</math></p> | A      |
| 1.17       | 4347a032 | <hr/> <p>How many <u>teaspoons</u> are equivalent to 44 tablespoons? (3 teaspoons = 1 tablespoon)</p> <p>A. 47</p> <p>B. 88</p> <p>C. 132</p> <p>D. 176</p>                                                    | C      |
| 1.18       | 3318d37b | <hr/> <p>A product costs 11.00 dollars per pound. What is the cost, in dollars, for 6 pounds of the product?</p>                                                                                               | 66     |
| 1.19       | 4837406c | <hr/> <p>An object travels at a constant speed of 6 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 24 centimeters?</p>                       | 4      |
| 1.20       | 94660ba8 | <hr/> <p>A participant in a bicycle race completes the race with an average speed of 24,816 yards per hour. What is this average speed, in <u>miles</u> per hour? (1 mile = 1,760 yards)</p>                   | 14.1   |

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| Question # | ID       |                                                                                                                                                                                                                                                                                                                           | Answer |
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| 1.21       | 551c52b9 | <p>Tilly earns <math>p</math> dollars for every <math>w</math> hours of work. Which expression represents the amount of money, in dollars, Tilly earns for <math>39w</math> hours of work?</p> <p>A. <math>39p</math></p> <p>B. <math>\frac{p}{39}</math></p> <p>C. <math>p + 39</math></p> <p>D. <math>p - 39</math></p> | A      |
| 1.22       | 7cd1c6db | <hr/> <p>An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?</p> <p>A. 9</p> <p>B. 96</p> <p>C. 120</p> <p>D. 972</p>                                                                           | A      |
| 1.23       | 86636d8f | <p>A customer spent \$27 to purchase oranges at \$3 per pound. How many pounds of oranges did the customer purchase?</p>                                                                                                                                                                                                  | 9      |